

Also taking the limit $u \rightarrow 0$, the single letter index is given by

$$f(t, 0; q) = \frac{t - q}{1 - q}. \quad (2.8)$$

In this case, the corresponding index is known as the “half-index” (of Neumann boundary condition) [14]. Due to an obvious symmetry between t and u , the limit $t \rightarrow 0$ is essentially same as $u \rightarrow 0$. However, in the analysis in the next subsection, these two limits look different, and lead to the equivalent result non-trivially.

Finally, if we set $u = q/t$, the resulting index is known as the flavored Schur index. The single letter index is now given by

$$f(t, q/t; q) = \frac{t + q/t - 2q}{1 - q}. \quad (2.9)$$

For the further specialization to $t = q^{1/2}$ (i.e., $u = q^{1/2}$), the index $I_N(q^{1/2}, q^{1/2}; q)$ is nothing but the original Schur index [3, 4].

We stress that all of them are obtained from the index (2.10) as special limits. The reduced index $I_N(t, u; q)$ is regarded as a two-parameter deformation of the Schur index $I_N(q^{1/2}, q^{1/2}; q)$. As mentioned in the introductory section, we refer to $I_N(t, u; q)$ as the deformed Schur index.

2.2 Exact evaluation of deformed Schur indices

In this subsection, we evaluate the matrix integral of the deformed Schur index exactly. When $v = p = 0$, we can rewrite the integral representation (2.4) as a more convenient form in terms of the q -Pochhammer symbol:

$$I_N(t, u; q) = \frac{1}{N!} \frac{(q; q)_\infty^N (tu; q)_\infty^N}{(t; q)_\infty^N (u; q)_\infty^N} \oint_{\mathbb{T}^N} \prod_{i=1}^N \frac{dx_i}{2\pi i x_i} \prod_{1 \leq i \neq j \leq N} \frac{(x_i/x_j; q)_\infty (tux_i/x_j; q)_\infty}{(tx_i/x_j; q)_\infty (ux_i/x_j; q)_\infty}. \quad (2.10)$$

This is a starting point of our analysis. The q -Pochhammer symbol is defined by

$$(x; q)_\infty = \prod_{k=0}^{\infty} (1 - xq^k), \quad (x; q)_n = \prod_{k=0}^{n-1} (1 - xq^k), \quad (x; q)_0 = 1, \quad (2.11)$$

and we have used an identity,

$$(qx; q)_\infty = \frac{(x; q)_\infty}{1 - x}, \quad (2.12)$$

to derive (2.10). A method to perform the integral (2.10) is simple. The computation consists of three steps.

In the first step, we recognize that the integrand of (2.10) includes a weight function of Macdonald polynomials of type A_{N-1} . In Appendix A, we review basics on the Macdonald polynomials of type A, based on [15, 16], for the reader’s convenience. The weight function of the Macdonald polynomials of type A_{N-1} is given by

$$w(x) = \prod_{1 \leq i \neq j \leq N} \frac{(x_i/x_j; q)_\infty}{(tx_i/x_j; q)_\infty}. \quad (2.13)$$